Water Management and Seasonal Wetlands in Aeronautical Infrastructure

De Almudena Urbieta

The current strategic context demands that the aeronautical industry move beyond simple waste management. The integration of rainwater harvesting systems is not an aesthetic goal, but an operational necessity for runoff management and compliance with discharge regulations. The creation of controlled rain ponds allows for the transformation of a potentially problematic water surplus into an asset for native biodiversity, reducing the load on industrial sewage systems.

The scientific framework of this volume is based on the dynamics of temporary wetlands. Unlike permanent bodies of water, rain ponds exhibit flood and desiccation cycles that favor specific taxa of local fauna, such as amphibians and aquatic macroinvertebrates, which have evolved to avoid permanent predators. These ecosystems act as biodiversity nodes, facilitating crucial ecosystem services such as water purification through natural phytoremediation processes.

Initial field analysis requires a comprehensive assessment of soil permeability and site topography. It is imperative to identify runoff reception basins within the factory to prevent the accumulation of hydrocarbon pollutants in areas designated for wildlife. Zoning must consider the safety distance to runways to mitigate bird strike risks, selecting plant species that are not attractive to large-scale avifauna.

The analysis phase is completed with the mapping of pre-existing native biodiversity. This is not about introducing exotic species, but about providing the necessary habitat for local species to colonize the space spontaneously. This biological monitoring process allows for the establishment of a baseline to evaluate the success of the mitigation measures implemented in the medium and long term.

Solutions engineering proposes the design of bioretention cells and sedimentation dikes. These structural elements allow for the attenuation of water surges, reducing flow velocity and facilitating the deposition of suspended solids before the water reaches the ponds intended for wildlife. The use of local materials and permeable geotextiles ensures slope stability without compromising natural infiltration.

In the second engineering phase, level control systems are installed using manual regulation gates. This allows for the management of the pond's hydroperiod according to seasonal biological needs and industrial safety protocols. Riparian vegetation is selected under strict criteria of resistance to water stress and nitrate filtration capacity, ensuring a resilient and low-maintenance system.

Success evaluation is quantified through biological and chemical Key Performance Indicators (KPIs). Monitoring water quality and counting amphibian egg masses are direct metrics of the return on investment in terms of biodiversity. A successful project demonstrates that industrial activity and the preservation of natural water cycles can coexist under rigorous technical management.

The action guide for the infrastructure manager begins with a water audit, followed by the design of a small-scale prototype. The scalability of these native biodiversity solutions allows for progressive implementation. The commitment to technical sustainability is not an option, but a pillar of industrial resilience against climate change and water resource scarcity.